



UNIVERSITY OF CALOOCAN CITY

Biglang Awa St. Corner 11th Ave. Catleya St. Caloocan City

Telephone Number: (02) 324 - 65 - 81

COMPUTER STUDIES DEPARTMENT

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY



“L.I.N.Y.A: Live Interactive Navigation of Your Appointment”

Bitancur, John Lawrence L.

Student

IT115 – IT Major Elective 4



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Abstract

Queue congestion in university offices remains a persistent operational challenge, often resulting in prolonged waiting times, inefficient service delivery, and weak identity verification processes. These issues negatively affect both administrative productivity and student satisfaction. This theoretical study introduces L.I.N.Y.A (Live Interactive Navigation of Your Appointment), a smart queuing system framework designed to enhance queue efficiency, personalization, and data security through the integration of face recognition technology, machine learning-based prediction, predictive notification mechanisms, and blockchain-based encryption.

The study examines relevant theoretical models in intelligent queuing systems, biometric authentication, predictive analytics, and decentralized data protection to establish a multidisciplinary foundation for the proposed framework. Using conceptual and framework-based analysis, the system architecture is evaluated in terms of operational workflow, identity validation accuracy, scalability potential, and security robustness. The model emphasizes proactive service preparation by forecasting queue demand patterns and dynamically informing users of their expected service time, thereby reducing uncertainty and physical crowding within university offices.

Furthermore, the proposed framework highlights the role of tamper-resistant blockchain storage in strengthening trust, auditability, and data integrity in appointment and identity records. By combining predictive intelligence with secure identity verification, L.I.N.Y.A demonstrates the potential to transform traditional first-come, first-served queuing into a more adaptive, user-centric service environment.

While the framework presents strong theoretical advantages, several limitations are acknowledged, including infrastructure and hardware requirements, computational overhead associated with real-time facial recognition and prediction models, potential privacy concerns, and the inherent uncertainty of machine learning forecasts. Future work is recommended to include prototype development, usability testing, real-world performance evaluation, and cost-benefit analysis within actual university environments.



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Introduction

Queue management plays a critical role in service-oriented environments, particularly in institutions where large populations request administrative support daily. University offices such as registrar, finance, admissions, and student services often experience heavy demand, resulting in overcrowding, inefficient queue flow, and user dissatisfaction. Traditional queue systems — whether manual or basic digital — primarily focus on number sequencing and display, offering limited intelligence, personalization, and predictive support.

Recent advances in artificial intelligence, biometric identification, and secure distributed data systems have enabled the development of smarter service platforms. These technologies allow systems not only to organize queues but also to recognize users, anticipate needs, automate notifications, and protect sensitive records. Integrating these capabilities into a unified queue management framework presents a promising direction for institutional service modernization.

Universities handle sensitive personal data and serve diverse user groups including students, faculty, and staff. Inefficient queues lead to time loss, administrative strain, and poor user experience. A smart queuing framework that combines biometric identification, predictive analytics, and secure storage directly addresses these operational weaknesses. This case was selected to theoretically examine how multiple intelligent technologies can be combined into a coherent queue management model tailored for academic environments.

Theoretically, this study bridges multiple computing domains — intelligent systems, biometric recognition, predictive modeling, and blockchain security — within a single service framework. It contributes to conceptual system design knowledge by demonstrating layered technology integration.

Practically, the study proposes a model that could improve office workflow, reduce congestion, enhance service readiness, and increase data protection. Even as a theoretical model, it provides design guidance for future smart campus systems.



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Statement of the Problem

Many current university queuing systems are limited to first-come-first-served numbering with static displays. These systems do not adapt to user history, cannot predict service needs, and rely on manual identity verification. As a result, returning users repeatedly submit the same information, service preparation is delayed, and queues move inefficiently.

Additionally, user data stored in centralized databases may be vulnerable to unauthorized modification or breaches if not strongly protected. There is a lack of integrated theoretical models that combine intelligent identification, predictive service support, proactive notification, and tamper-resistant storage in queue management systems.

Objectives of the Study

- **General Objective**

- To develop and analyze a theoretical framework for a smart university queuing system that integrates biometric recognition, machine learning prediction, predictive notifications, and blockchain encryption to improve efficiency, security, and service organization.

- **Specific Objectives**

- To examine intelligent queuing theories and service flow principles that support efficient queue management.
- To analyze enabling technologies including biometric face recognition for repeat-user identification and machine learning models for service intent forecasting.
- To evaluate predictive notification strategies such as SMS and email alerts within smart queue environments.
- To assess blockchain-based frameworks for data integrity, encryption, and secure record management.
- To design and propose an integrated smart queue architecture that identifies strengths, theoretical risks, and future readiness for academic offices.

Significance of the Study

This study contributes a cross-domain conceptual framework that unifies biometric authentication, predictive analytics, and blockchain security within queue management theory. It expands theoretical discussion on intelligent service automation and secure administrative system design.

The study supports academic research in artificial intelligence applications in administration, smart campus systems, predictive service modeling, biometric identity



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frameworks, secure distributed databases, and human–computer interaction in public service systems. These areas collectively strengthen the development of intelligent, efficient, and user-centered digital service environments within institutional settings.

Future research may extend this framework through prototype development and usability testing, predictive model accuracy measurement, biometric performance benchmarking, blockchain storage efficiency analysis, user acceptance and trust studies, and comparative queue performance evaluation. Such directions can further validate system effectiveness and enhance the scalability, security, and overall performance of next-generation public service technologies.

Scope and Limitations

- **Scope**

- Conceptual smart queuing architecture integrates real-time queue number generation, tracking, and analytics within university administrative office environments.
- Biometric identification is enabled through face recognition to accurately recognize and manage returning users.
- Machine learning–based prediction is utilized to forecast service demand and optimize queue flow and preparation.
- Predictive notification models deliver timely SMS and email updates to keep users informed of their queue status.
- Blockchain-encrypted storage secures user templates and records while supporting theory-driven system design and evaluation.

- **Limitations**

- No empirical validation was conducted, as the study does not include live system deployment or real biometric dataset evaluation.
- Machine learning performance is examined only at the conceptual level without experimental training or accuracy measurement.
- Blockchain efficiency issues such as overhead and latency are discussed but not empirically measured.
- System usability remains unverified because no user testing or human–computer interaction evaluation was performed.
- Implementation feasibility is addressed theoretically, with conclusions based on secondary sources and integration complexity not fully engineered.



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Proposed by:

Bitancur, John Lawrence L.

PROF. FLOR, ROMEO RAPHAEL JOHN
IT 115 – IT MAJOR ELECTICE 4